

Yohei Ohata

National Institute of Information and Communications Technology
Center for Information and Neural Networks
1-4, Yamadaoka, Suita, Osaka, Japan

yohei.ohata.ee@nict.go.jp

Education

The University of Tokyo

April 2022 – March 2024

M.S. in Human Factors Engineering
Department of Human and Engineered Environmental Studies
Cumulative GPA: 4.0/4.0

Thesis: Identifying Different Thought Processes for Each Learner Using Eye-Tracking and Hidden Markov Models

Tokyo University of Science

April 2018 – March 2022

B.E. in Electrical and Electronic Engineering
Department of Applied Electronics
Cumulative GPA: 3.5/4.0

Thesis: A Method of Designing a Hilbert Transformer with Transmission Zeros at Specified Positions

Honors and Awards

- 2024 Dean's Award for Best Thesis and Academic Performance (ranked 1st of 48)
 Department of Human and Engineered Environmental Studies, The University of Tokyo
- 2021 IEEE Circuits and Systems Society Japan Joint Chapter Best Student Award
 for top presentation of the year

Research Experience

Research Engineer

January 2026 – Present

National Institute of Information and Communications Technology
Center for Information and Neural Networks
Computational Neuroscience & Decision Making Laboratory
Supervisor: Masahiko Haruno

- Investigate the extent to which attention mechanisms in Transformer architectures share functional and representational properties with hippocampal computation, including where the two systems diverge
- Train a single-layer, attention-only Transformer on continuous sequences generated by a Gaussian Hidden Markov Model to identify the conditions under which the model can learn the underlying states

Cooperative Visiting Researcher

August 2025 – Present

The University of Tokyo
Graduate School of Information Science and Technology
Amano & Nakayama Lab
Supervisor: Ayumu Yamashita, Kaoru Amano

- Analyze public fMRI datasets, including AX-CPT, Cued Task Switching, Sternberg, and Stroop tasks, to identify task-general and task-specific neural substrates of cognitive control at both inter- and inner-subject levels
- Preprocess datasets and synchronize inter-subject time series using BrainSync in MATLAB, applying tensor decomposition to extract multiple network components across spatial, temporal, and subject modes

- Identify task-general and task-specific neural networks by evaluating inter-task correlations across spatial modes
- Examine networks associated with inter-subject variability in cognitive control performance using subject modes and conditional randomization testing
- Identify neural substrates underlying inner-subject sustained attention states using temporal modes and generalized linear mixed models

Research Assistant

April 2022 – March 2024

The University of Tokyo
Department of Human and Engineered Environmental Studies
Inovative Learning Creation Lab
Supervisor: Satori Hachisuka, Shin'ichi Warisawa

- Designed experimental protocols for both adult and elementary school participants, including the development of a web-based UI and logging system using HTML/CSS/Javascript
- Managed high-fidelity eye-tracking data collection and processing using Tobii Pro 3 and Tobii Pro Lab
- Modeled the temporal dynamics of scan paths using Hidden Markov Models implemented in Python

Research Assistant

May 2021 – December 2021

Tokyo University of Science
Department of Applied Electronics
Signal Processing Lab
Supervisor: Naoyuki Aikawa

- Proposed a Composite Hilbert Transformer that integrates a pre-normalized notch filter to suppress noise at specific frequencies
- Verified significant SNR improvements over conventional Band Pass Hilbert Transformer through extensive simulations in MATLAB

Publications

Journal Articles

Ohata, Y., Yamashita, A., & Amano, K. (in preparation). Neural Substrates of Cognitive Control and Their Individual Variability.

Ohata, Y., & Haruno, M. (in preparation). Convergent and Divergent Computational Principles of Context Processing in Transformer Attention and the Hippocampal Formation.

Ohata, Y., Hachisuka, S., Kurita, K., & Warisawa, S. (submitted). Identifying Different Thought Processes for Each Learner Using Eye-Tracking and Hidden Markov Models. *Frontiers in Psychology*.

Peer-Reviewed Conference Proceedings

Ohata, Y., Hachisuka, S., Kurita, K., & Warisawa, S. (2024). [Oral]. Visualizing and Revealing the Difference of Learner's Thought Process and Question Solving Method. *In Proceedings of International Conference on Information and Communication Technology*.

Ohata, Y., Takao, K., Natori, T., & Aikawa, N. (2021). [Oral]. A Method of Designing a Hilbert Transformer with Transmission Zeros at Specified Positions. *ICICE Technical Report*.

Conference Presentations

Ohata, Y., Hachisuka, S., Kurita, K., & Warisawa, S. (2023). [Poster]. Analysis of How Learners Learn Using Graphic Questions. *The 42nd Annual Conference of Japanese Society for Educational Technology*.

Ohata, Y., Hachisuka, S., Kurita, K., & Warisawa, S. (2022). [Poster]. Identifying Perceptual Factors that Influence Impression of a Teacher in a Classroom. *The 41st Annual Conference of Japanese Society for Educational Technology*.

Professional Experience

Research Engineer

April 2024 – December 2025

Hitachi, Ltd., Digital Innovation R&D

- Developed concepts and business use cases, and defined the corresponding functional requirements and methodologies
- Applied LLMs in service development and established evaluation criteria for LLM verification
- Filed 2 patent applications

Teaching Experience

Teaching Assistant

January 2026 – Present

National Institute of Information and Communications Technology

Center for Information and Neural Networks

Course: Computational Neuroscience & Decision Making Laboratory AI Study Club

- Deliver technical tutorials on Transformer computational principles for 20+ lab members, covering multi-head attention, query-key/value-output circuits, and induction heads through a mechanistic interpretability lens
- Introduce modern machine learning tools and implementation practices, including PyTorch, JAX, and uv, to strengthen lab members' technical workflows
- Curate and present 10+ landmark NeuroAI papers from the past five years, facilitating cross-disciplinary discussion between neuroscience and artificial intelligence

Teaching Assistant

April 2023 – June 2023

The University of Tokyo

Department of Human and Engineered Environmental Studies

Course: Sensing and Signal Processing

- Guided students in building signal processing systems using LabVIEW

Teacher

April 2019 – March 2022

Gakuho Seminar

- Taught Mathematics, Physics, Chemistry, and English to students in Years 10–18.
- Created and managed individualized study plans to support academic progress.
- Successfully guided over 30 students in passing their target school entrance examinations.

References

Masahiko Haruno, Ph.D.,

Professor

National Institute of Information and Communications Technology

Email: mharuno@nict.go.jp

Ayumu Yamashita, Ph.D.,

Associate Professor

Kobe University

Advanced Telecommunications Research Institute International

Email: ayumu@atr.jp

Satori Hachisuka, Ph.D.,

Associate Professor

The University of Tokyo

Email: hachisuka@edu.k.u-tokyo